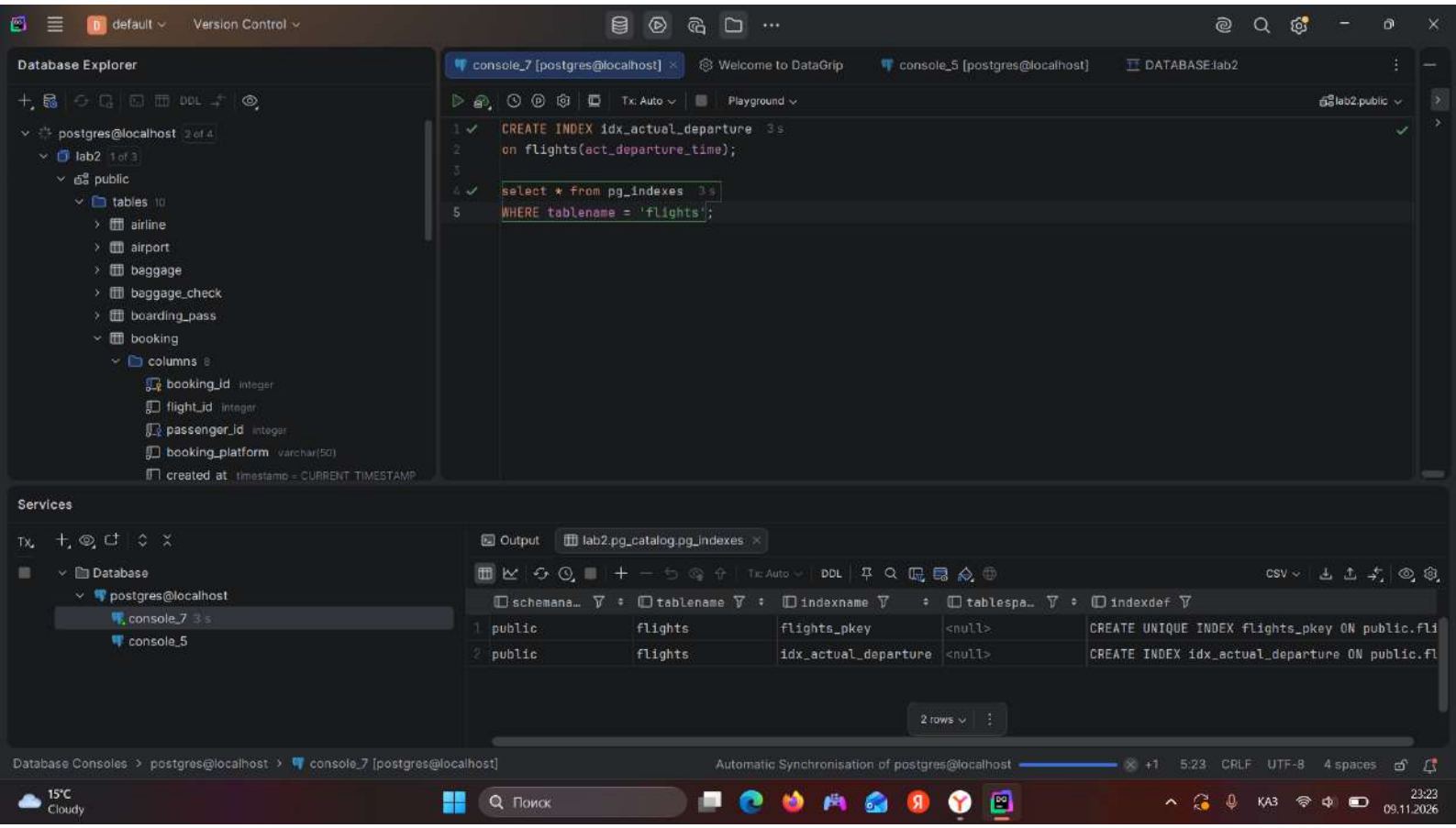
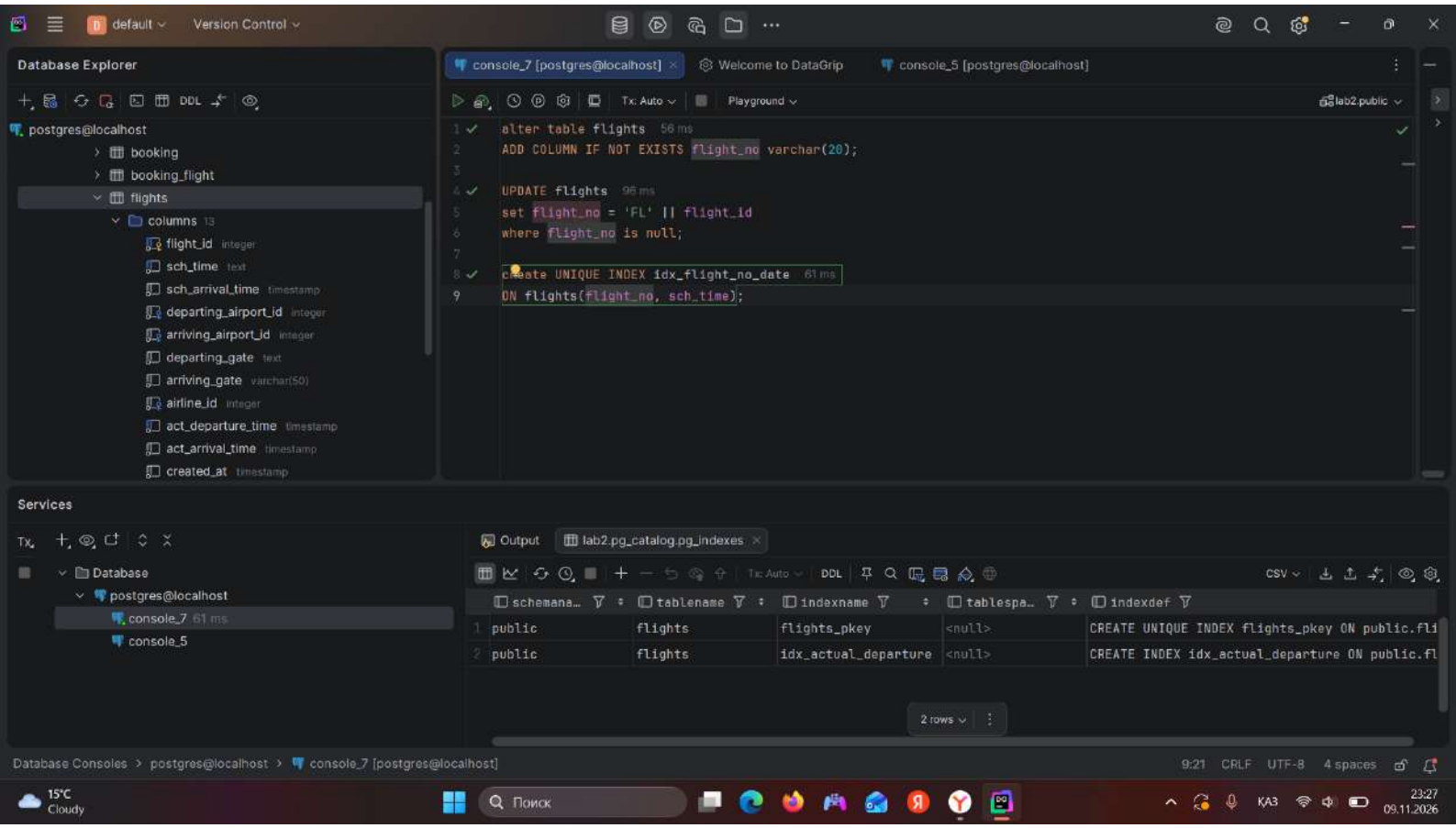
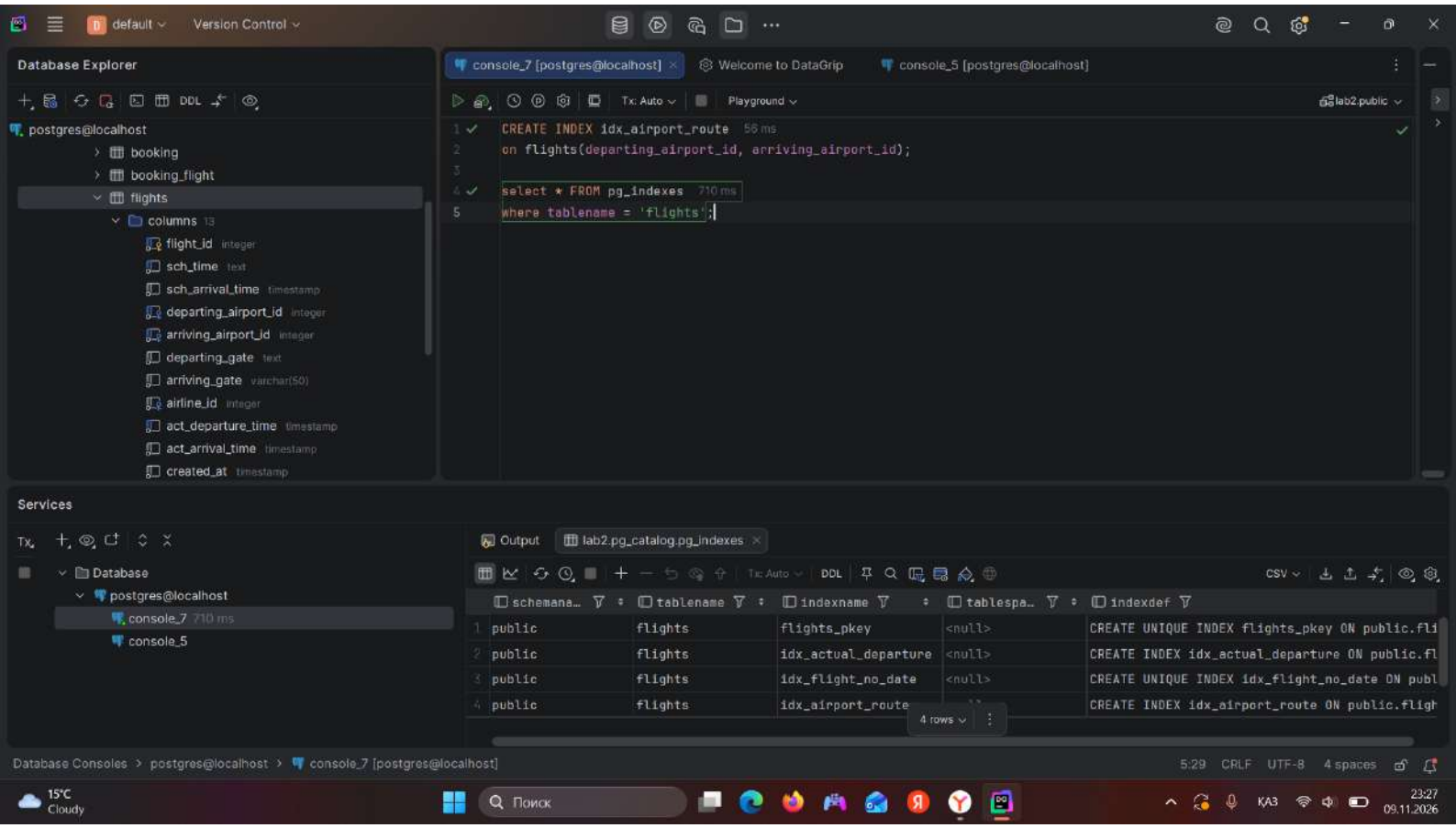
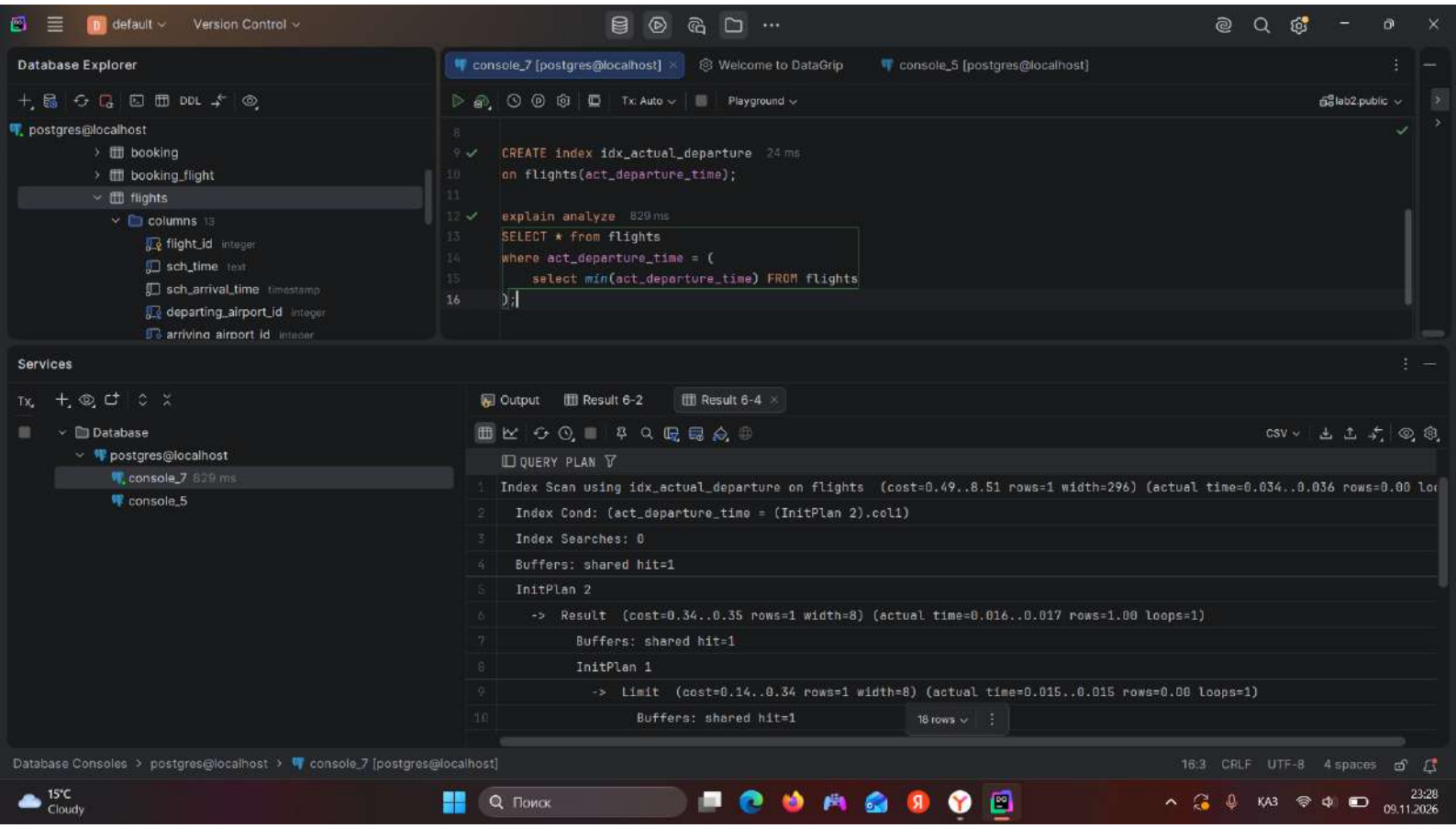
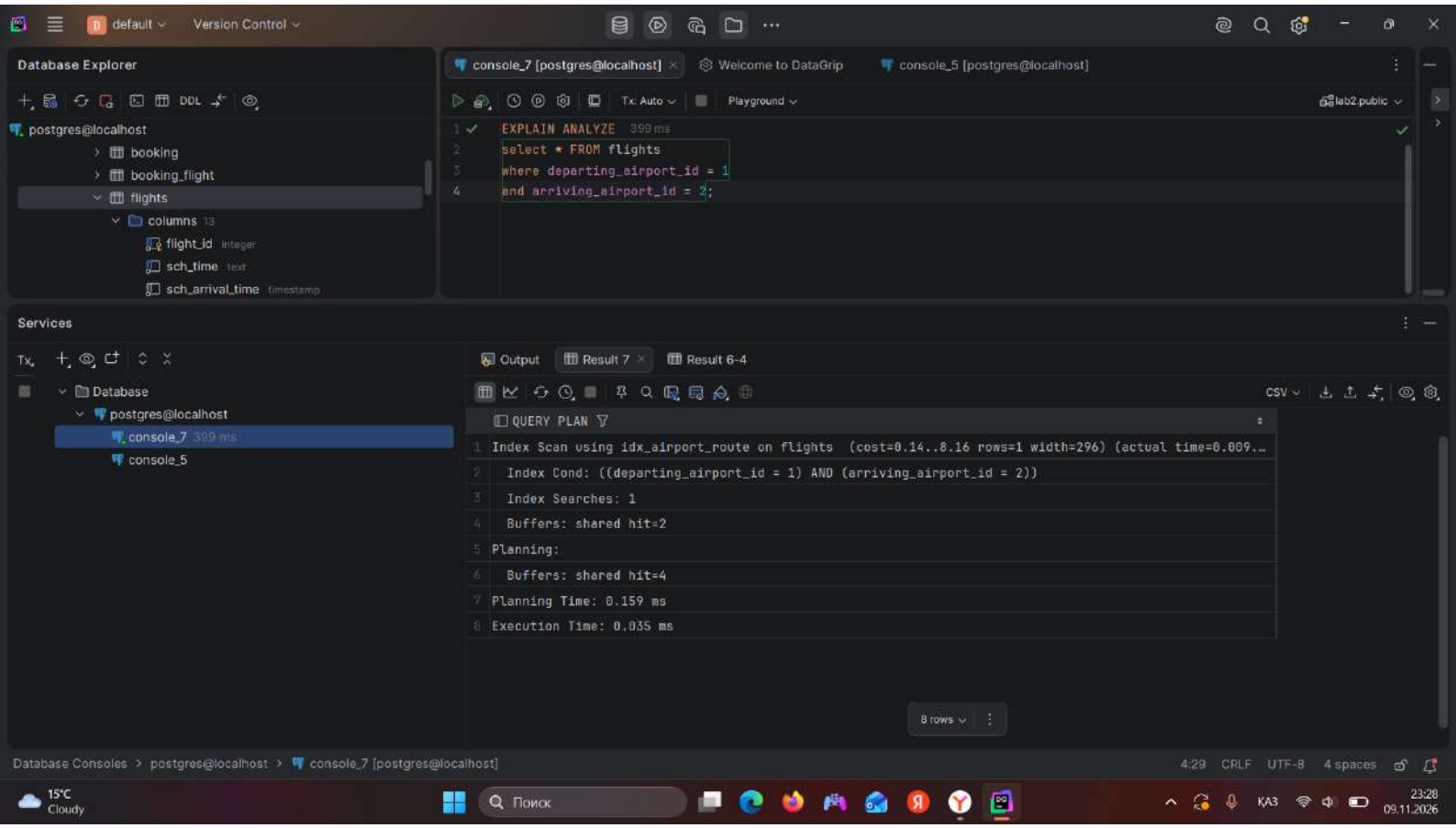
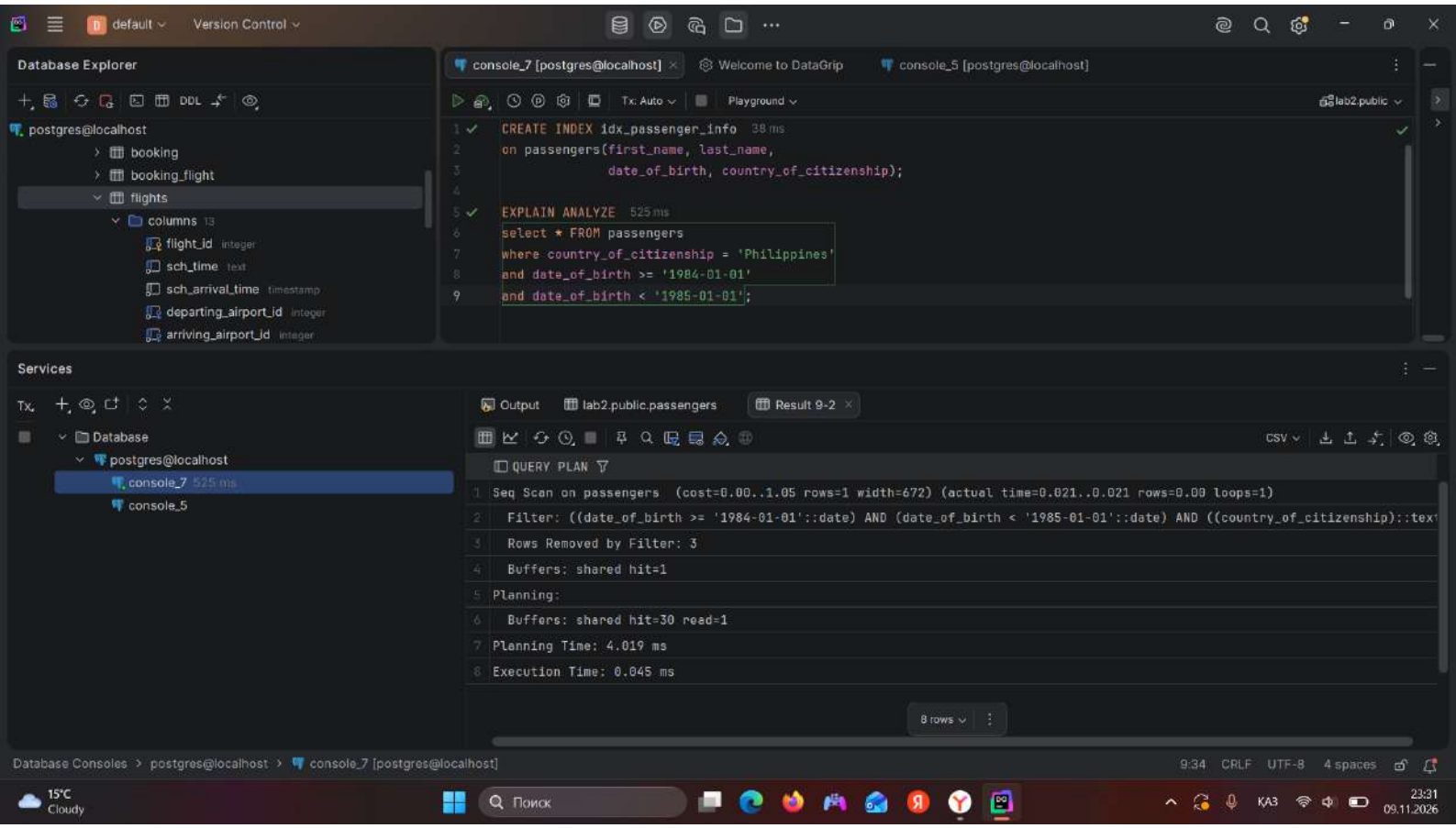

default
Version Control
Database Explorer
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
DATABASE:lab2
Tx: Auto
Playground
lab2.public
postgres@localhost 2 of 4
lab2 1 of 3
public
tables 10
airline
airport
baggage
baggage_check
boarding_pass
booking
columns 8
booking_id integer
flight_id integer
passenger_id integer
booking_platform varchar(50)
created_at timestamp = CURRENT_TIMESTAMP
1 CREATE INDEX idx_actual_departure 3 s
2 on flights(act_departure_time);
3
4 select * from pg_indexes 3 s
5 WHERE tablename = 'flights';
Services
Database
postgres@localhost
console_7 3 s
console_5
Output
lab2.pg_catalog.pg_indexes
schemana…
tablename
indexname
tablespa…
indexdef
1 public flights flights_pkey <null> CREATE UNIQUE INDEX flights_pkey ON public.fli
2 public flights idx_actual_departure <null> CREATE INDEX idx_actual_departure ON public.fl
2 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
Automatic Synchronisation of postgres@localhost
5:23
CRLF
UTF-8
4 spaces
15°C
Cloudy
Поиск
ҚАЗ
23:23
09.11.2026

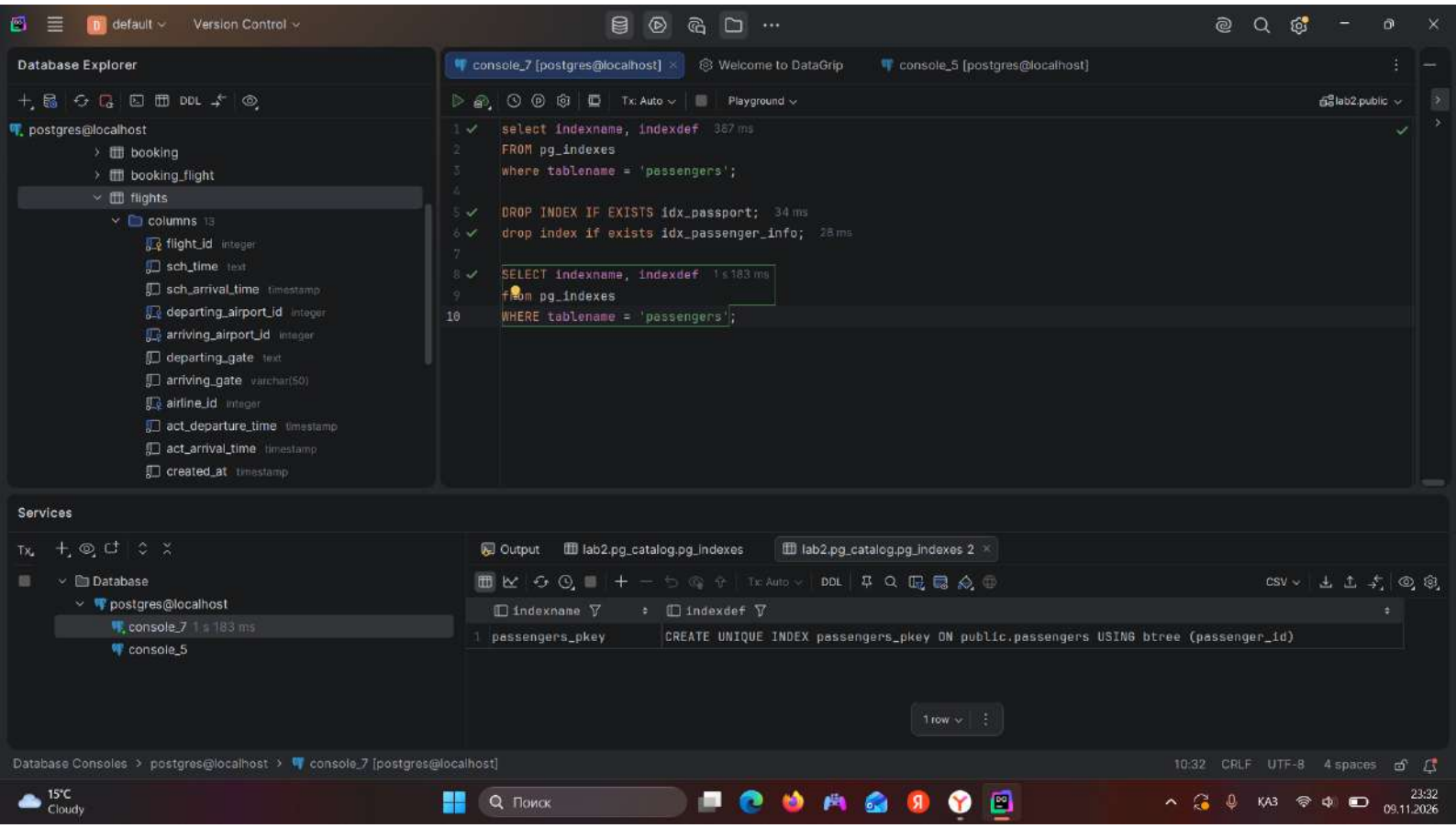

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
created_at timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 alter table flights 56 ms
2 ADD COLUMN IF NOT EXISTS flight_no varchar(20);
3
4 UPDATE flights 96 ms
5 set flight_no = 'FL' || flight_id
6 where flight_no is null;
7
8 create UNIQUE INDEX idx_flight_no_date 61 ms
9 ON flights(flight_no, sch_time);
Services
Database
postgres@localhost
console_7 61 ms
console_5
Output
lab2.pg_catalog.pg_indexes
Tx: Auto
DDL
CSV
schemana…
tablename
indexname
tablespa…
indexdef
1 public flights flights_pkey <null> CREATE UNIQUE INDEX flights_pkey ON public.fli
2 public flights idx_actual_departure <null> CREATE INDEX idx_actual_departure ON public.fl
2 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
9:21 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:27
09.11.2026

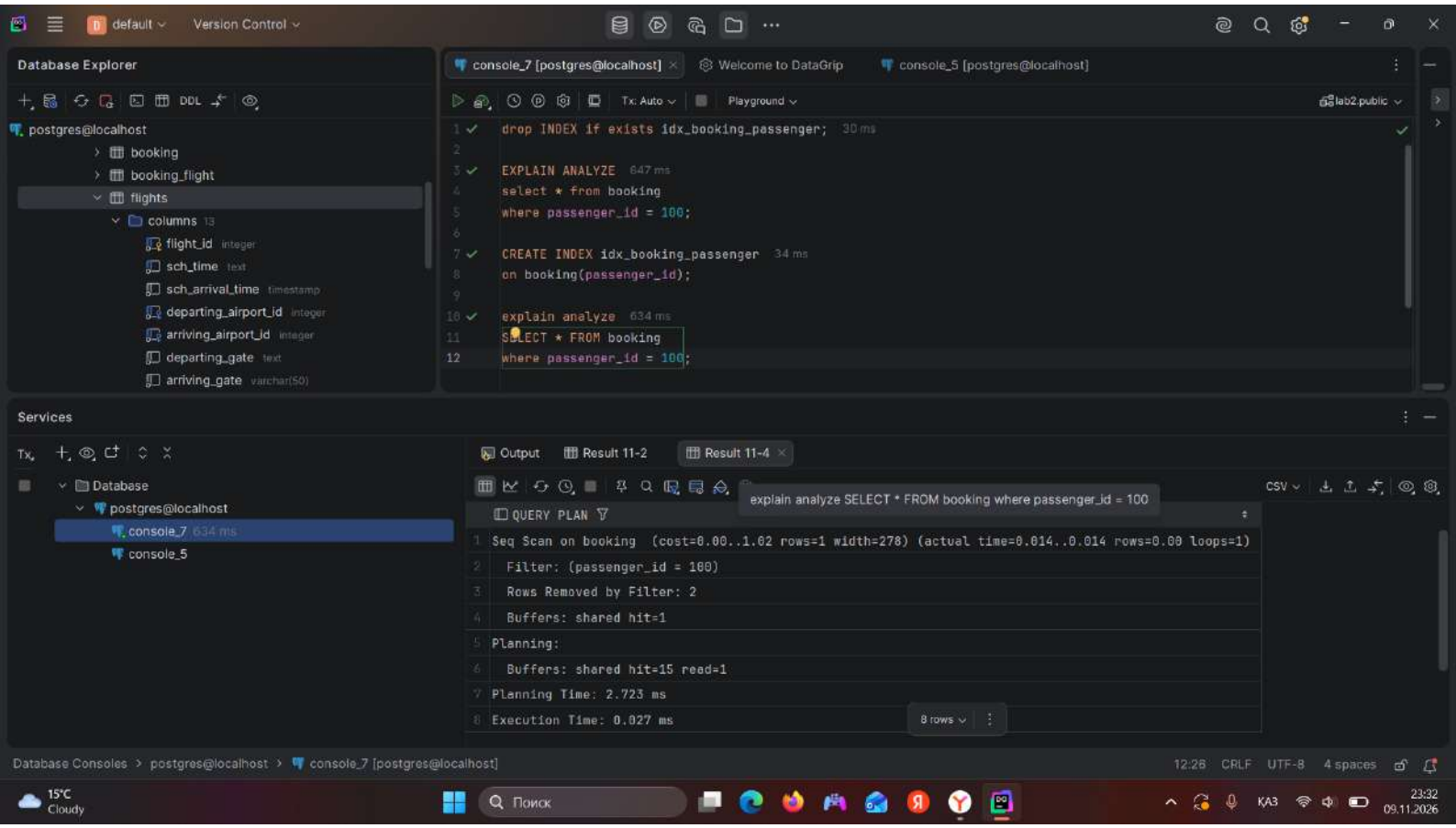

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
created_at timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 CREATE INDEX idx_airport_route 56 ms
2 on flights(departing_airport_id, arriving_airport_id);
3
4 select * FROM pg_indexes 710 ms
5 where tablename = 'flights';
Services
Database
postgres@localhost
console_7 710 ms
console_5
Output
lab2.pg_catalog.pg_indexes
CSV
schemana… tablename indexname tablespa… indexdef
1 public flights flights_pkey <null> CREATE UNIQUE INDEX flights_pkey ON public.fli
2 public flights idx_actual_departure <null> CREATE INDEX idx_actual_departure ON public.fl
3 public flights idx_flight_no_date <null> CREATE UNIQUE INDEX idx_flight_no_date ON publ
4 public flights idx_airport_route CREATE INDEX idx_airport_route ON public.fligh
4 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
5:29 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:27
09.11.2026

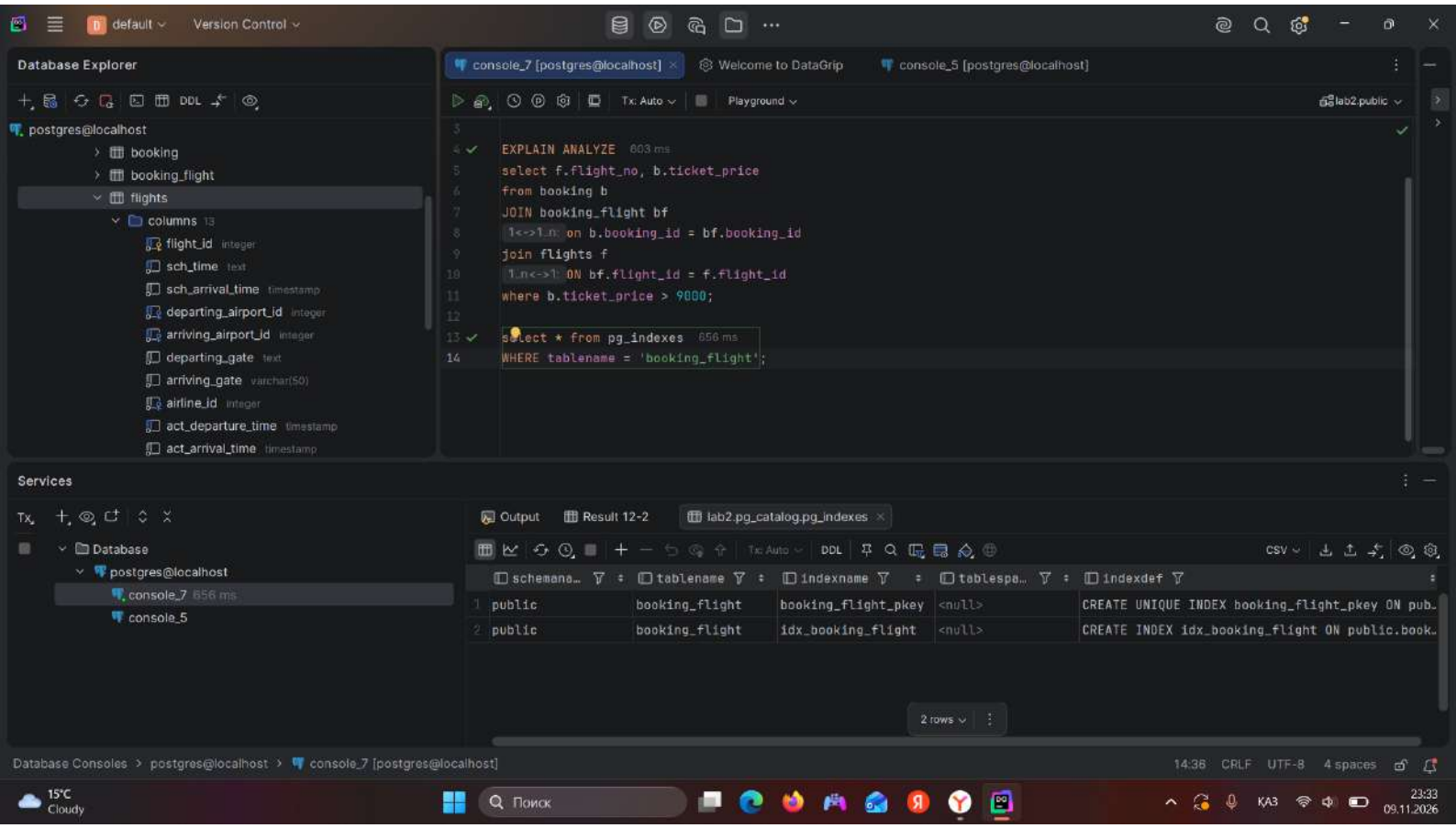

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
CREATE index idx_actual_departure 24 ms
on flights(act_departure_time);
explain analyze 829 ms
SELECT * from flights
where act_departure_time = (
select min(act_departure_time) FROM flights
);
Services
Database
postgres@localhost
console_7 829 ms
console_5
Output
Result 6-2
Result 6-4
CSV
QUERY PLAN
Index Scan using idx_actual_departure on flights (cost=0.49..8.51 rows=1 width=296) (actual time=0.034..0.036 rows=0.00 loo
Index Cond: (act_departure_time = (InitPlan 2).col1)
Index Searches: 0
Buffers: shared hit=1
InitPlan 2
-> Result (cost=0.34..0.35 rows=1 width=8) (actual time=0.016..0.017 rows=1.00 loops=1)
Buffers: shared hit=1
InitPlan 1
-> Limit (cost=0.14..0.34 rows=1 width=8) (actual time=0.015..0.015 rows=0.00 loops=1)
Buffers: shared hit=1
18 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
16:3 CRLF UTF-8 4 spaces
15°C
Cloudy
Поиск
ҚАЗ
23:28
09.11.2026

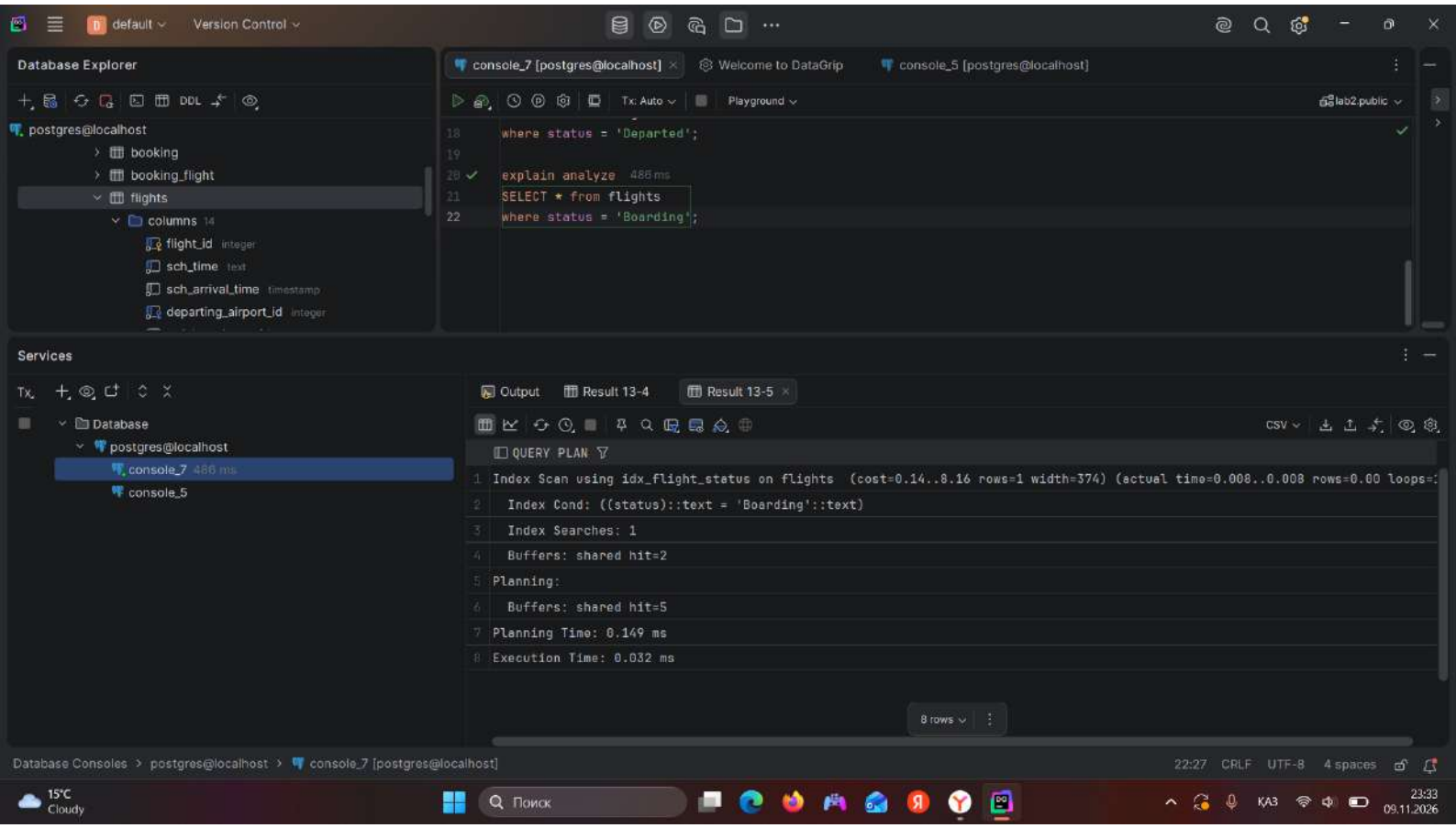

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
EXPLAIN ANALYZE 399 ms
select * FROM flights
where departing_airport_id = 1
and arriving_airport_id = 2;
Services
Database
postgres@localhost
console_7 399 ms
console_5
Output
Result 7
Result 6-4
CSV
QUERY PLAN
Index Scan using idx_airport_route on flights (cost=0.14..8.16 rows=1 width=296) (actual time=0.009...
Index Cond: ((departing_airport_id = 1) AND (arriving_airport_id = 2))
Index Searches: 1
Buffers: shared hit=2
Planning:
Buffers: shared hit=4
Planning Time: 0.159 ms
Execution Time: 0.035 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
4:29 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:28
09.11.2026

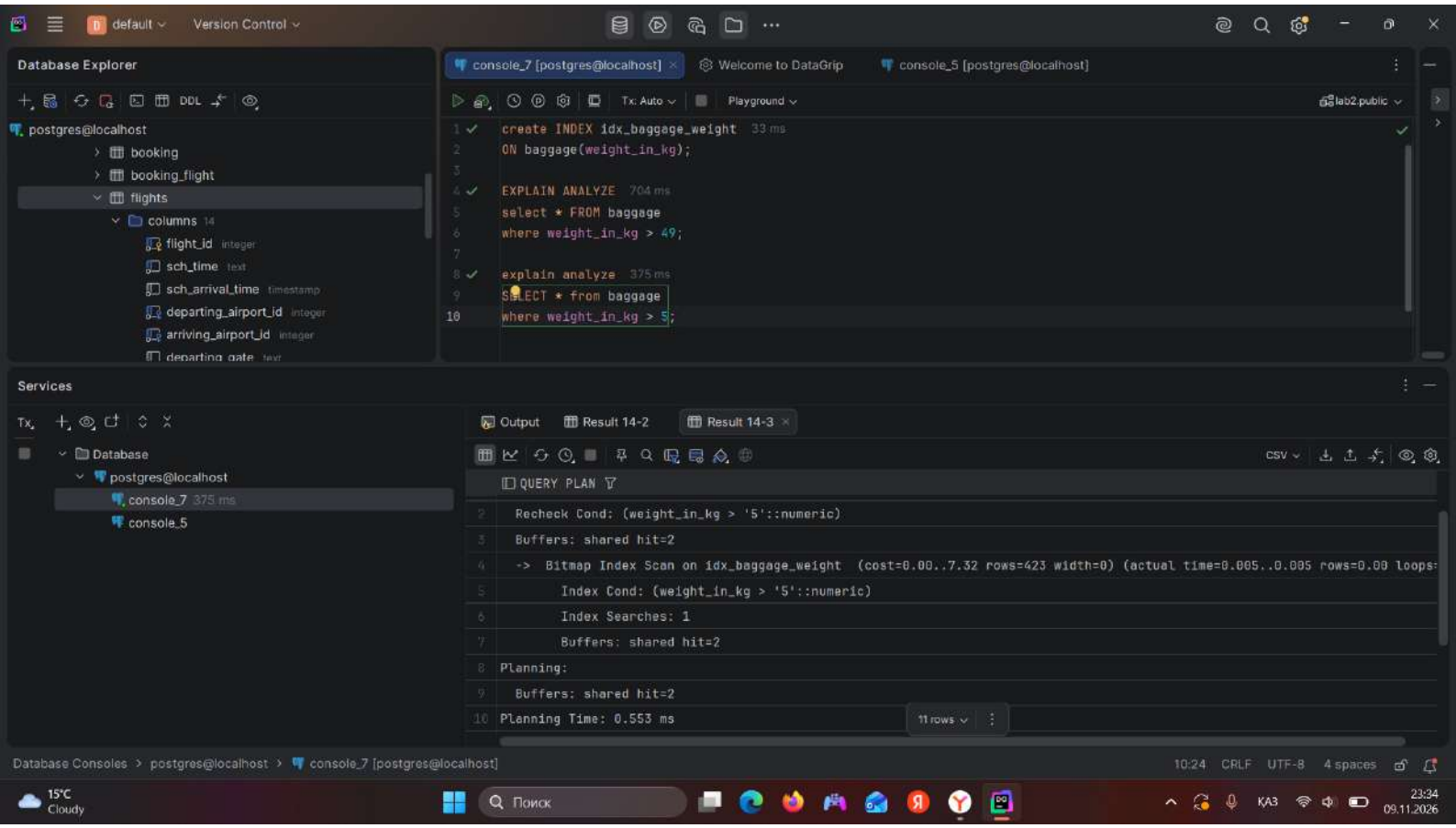

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
CREATE INDEX idx_passenger_info 38 ms
on passengers(first_name, last_name,
date_of_birth, country_of_citizenship);
EXPLAIN ANALYZE 525 ms
select * FROM passengers
where country_of_citizenship = 'Philippines'
and date_of_birth >= '1984-01-01'
and date_of_birth < '1985-01-01';
Services
Database
postgres@localhost
console_7 525 ms
console_5
Output
lab2.public.passengers
Result 9-2
CSV
QUERY PLAN
Seq Scan on passengers (cost=0.00..1.05 rows=1 width=672) (actual time=0.021..0.021 rows=0.00 loops=1)
Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::tex
Rows Removed by Filter: 3
Buffers: shared hit=1
Planning:
Buffers: shared hit=30 read=1
Planning Time: 4.019 ms
Execution Time: 0.045 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
9:34 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:31
09.11.2026

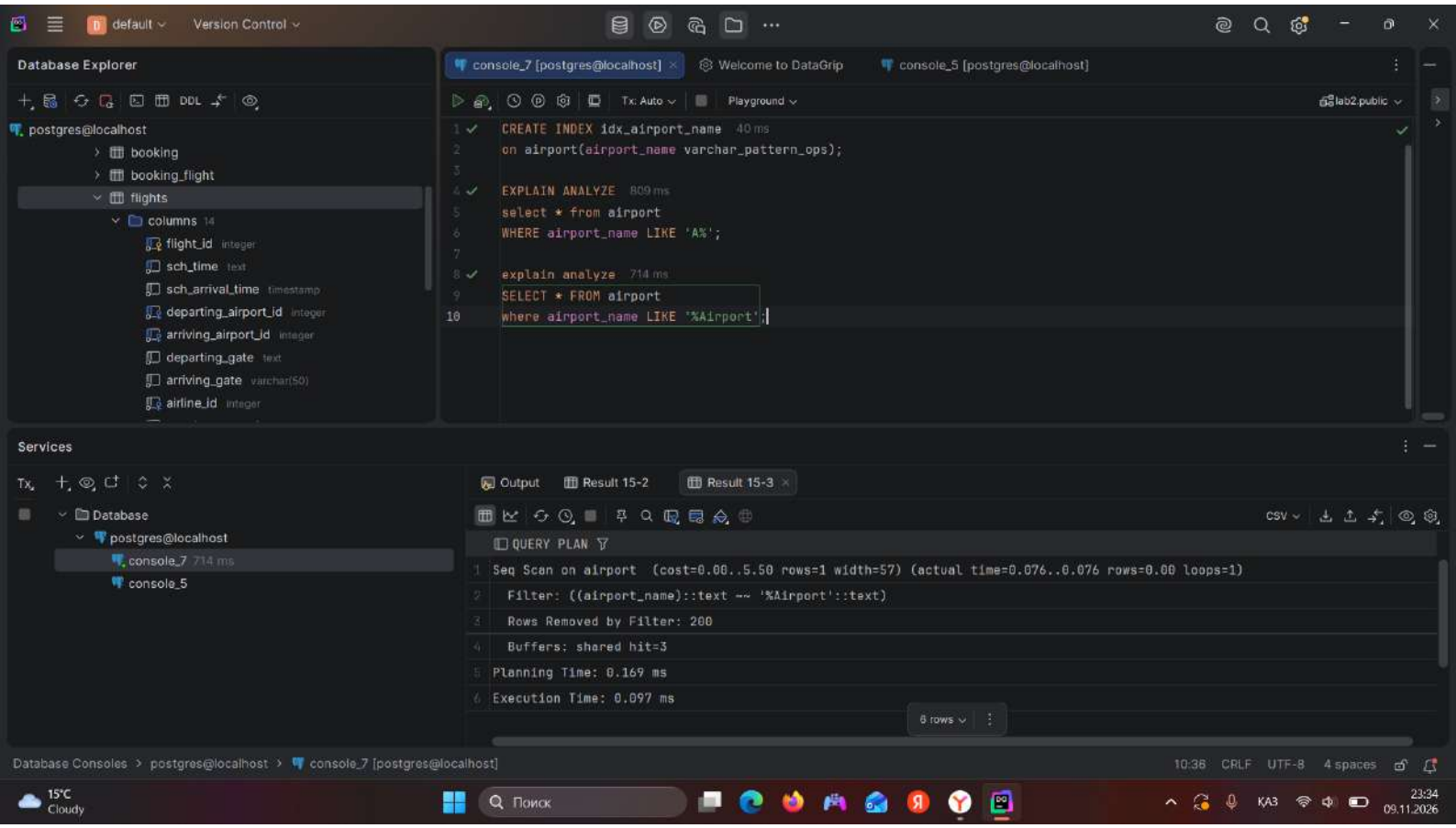

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
created_at timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 select indexname, indexdef 387 ms
2 FROM pg_indexes
3 where tablename = 'passengers';
4
5 DROP INDEX IF EXISTS idx_passport; 34 ms
6 drop index if exists idx_passenger_info; 28 ms
7
8 SELECT indexname, indexdef 1 s 183 ms
9 from pg_indexes
10 WHERE tablename = 'passengers';
Services
Database
postgres@localhost
console_7 1 s 183 ms
console_5
Output
lab2.pg_catalog.pg_indexes
lab2.pg_catalog.pg_indexes 2
Tx: Auto
DDL
CSV
indexname
indexdef
1 passengers_pkey
CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
1 row
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
10:32 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:32
09.11.2026

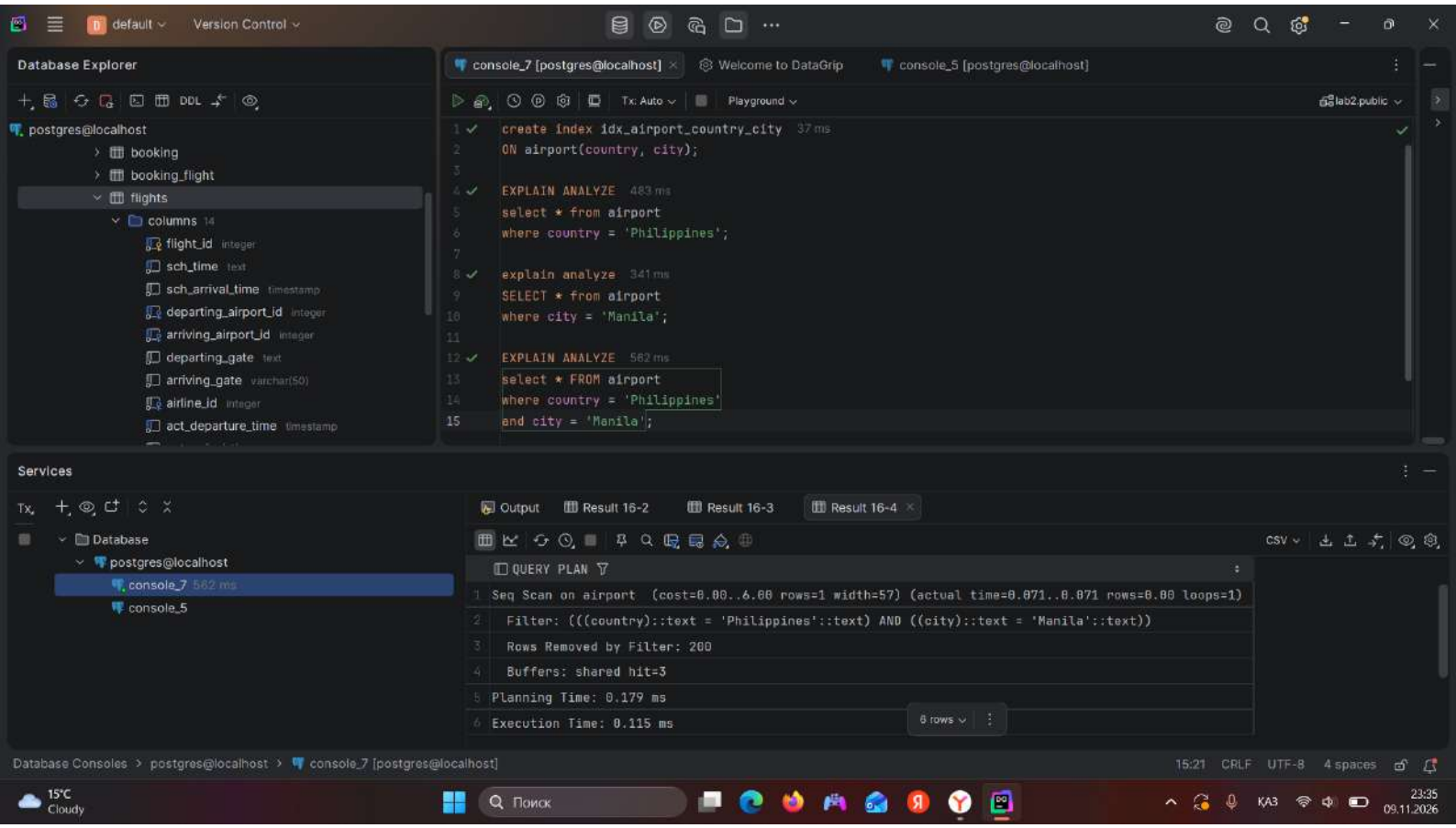

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 drop INDEX if exists idx_booking_passenger; 30 ms
3 EXPLAIN ANALYZE 647 ms
4 select * from booking
5 where passenger_id = 100;
7 CREATE INDEX idx_booking_passenger 34 ms
8 on booking(passenger_id);
10 explain analyze 634 ms
11 SELECT * FROM booking
12 where passenger_id = 100;
Services
Database
postgres@localhost
console_7 634 ms
console_5
Output
Result 11-2
Result 11-4
CSV
explain analyze SELECT * FROM booking where passenger_id = 100
QUERY PLAN
1 Seq Scan on booking (cost=0.00..1.02 rows=1 width=278) (actual time=0.014..0.014 rows=0 loops=1)
2 Filter: (passenger_id = 100)
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=15 read=1
7 Planning Time: 2.723 ms
8 Execution Time: 0.027 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
12:28 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:32
09.11.2026

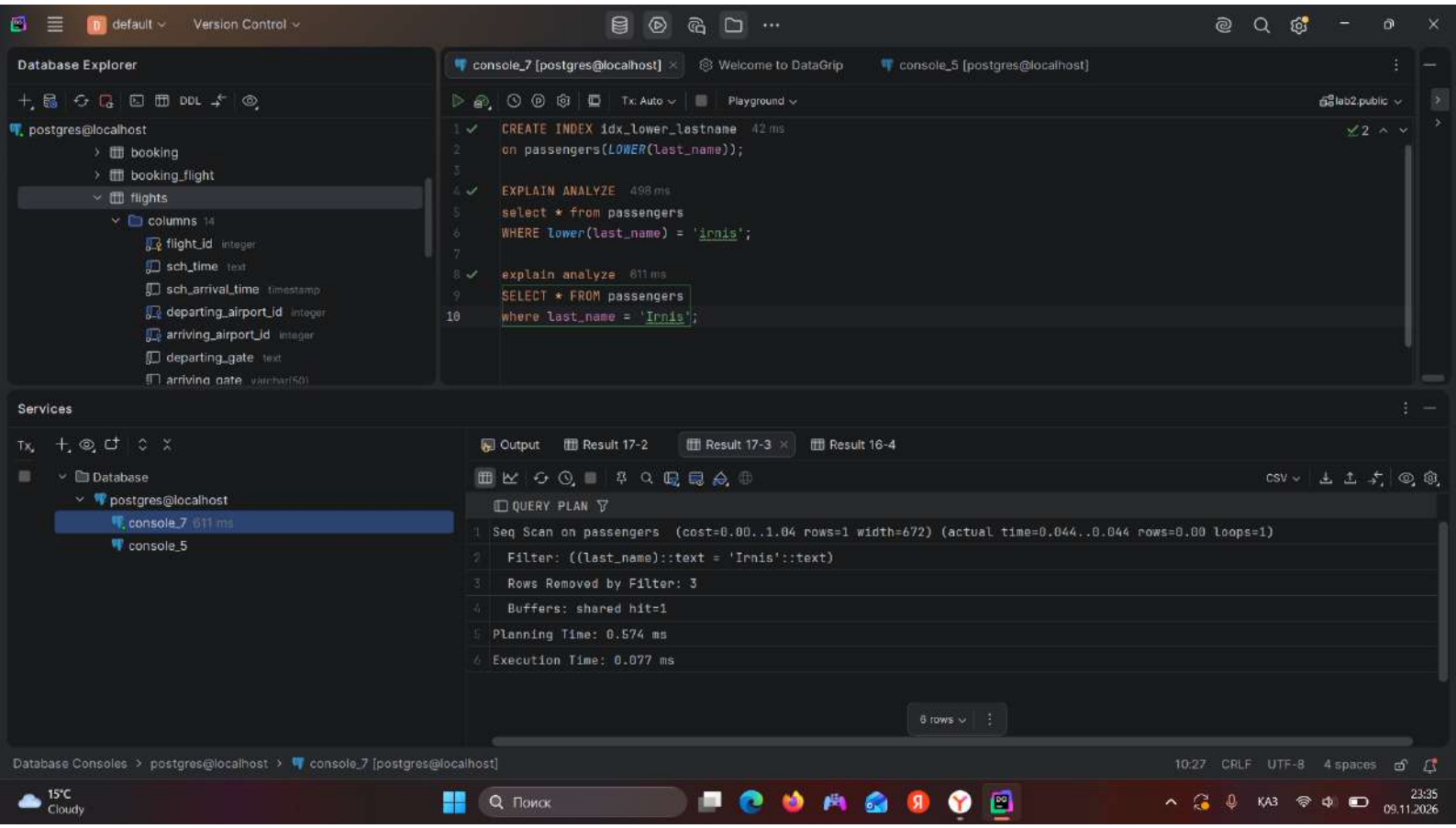

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 13
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
EXPLAIN ANALYZE 603 ms
select f.flight_no, b.ticket_price
from booking b
JOIN booking_flight bf
1<->1..n: on b.booking_id = bf.booking_id
join flights f
1..n<->1: ON bf.flight_id = f.flight_id
where b.ticket_price > 9000;
select * from pg_indexes 656 ms
WHERE tablename = 'booking_flight';
Services
Database
postgres@localhost
console_7 656 ms
console_5
Output
Result 12-2
lab2.pg_catalog.pg_indexes
CSV
schemana…
tablename
indexname
tablespa…
indexdef
1 public booking_flight booking_flight_pkey <null> CREATE UNIQUE INDEX booking_flight_pkey ON pub…
2 public booking_flight idx_booking_flight <null> CREATE INDEX idx_booking_flight ON public.book…
2 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
14:36 CRLF UTF-8 4 spaces
15°C
Cloudy
Поиск
ҚАЗ
23:33
09.11.2026

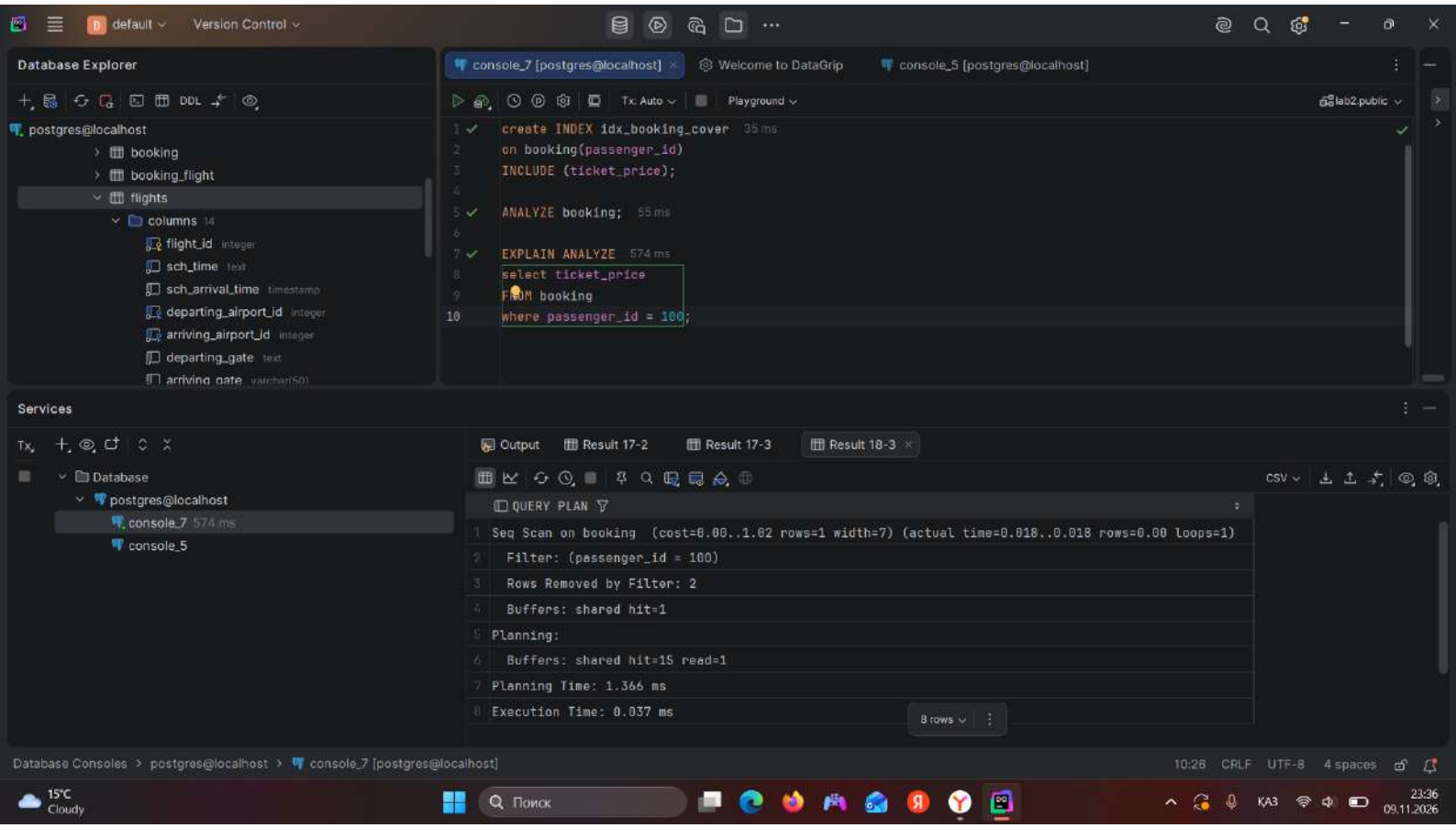

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
where status = 'Departed';
explain analyze 486 ms
SELECT * from flights
where status = 'Boarding';
Services
Database
postgres@localhost
console_7 486 ms
console_5
Output
Result 13-4
Result 13-5
CSV
QUERY PLAN
Index Scan using idx_flight_status on flights (cost=0.14..8.16 rows=1 width=374) (actual time=0.008..0.008 rows=0.00 loops=1
Index Cond: ((status)::text = 'Boarding'::text)
Index Searches: 1
Buffers: shared hit=2
Planning:
Buffers: shared hit=5
Planning Time: 0.149 ms
Execution Time: 0.032 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
22:27 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:33
09.11.2026

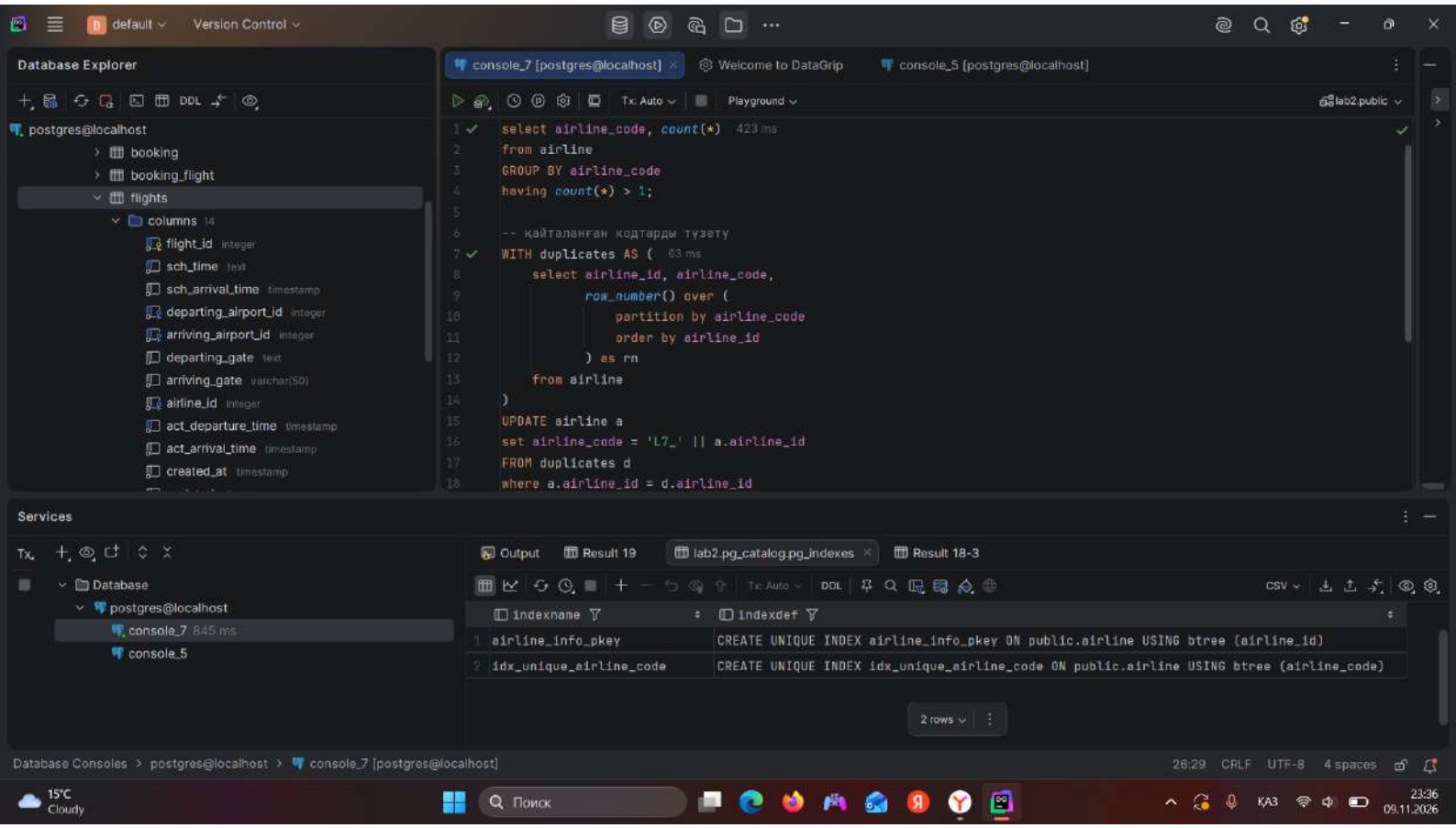

default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 create INDEX idx_baggage_weight 33 ms
2 ON baggage(weight_in_kg);
3
4 EXPLAIN ANALYZE 704 ms
5 select * FROM baggage
6 where weight_in_kg > 49;
7
8 explain analyze 375 ms
9 SELECT * from baggage
10 where weight_in_kg > 5;
Services
Database
postgres@localhost
console_7 375 ms
console_5
Output
Result 14-2
Result 14-3
CSV
QUERY PLAN
2 Recheck Cond: (weight_in_kg > '5'::numeric)
3 Buffers: shared hit=2
4 -> Bitmap Index Scan on idx_baggage_weight (cost=0.00..7.32 rows=423 width=0) (actual time=0.005..0.005 rows=0 loops=
5 Index Cond: (weight_in_kg > '5'::numeric)
6 Index Searches: 1
7 Buffers: shared hit=2
8 Planning:
9 Buffers: shared hit=2
10 Planning Time: 0.553 ms
11 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
10:24 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:34
09.11.2026


default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 CREATE INDEX idx_airport_name 40 ms
2 on airport(airport_name varchar_pattern_ops);
3
4 EXPLAIN ANALYZE 809 ms
5 select * from airport
6 WHERE airport_name LIKE 'A%';
7
8 explain analyze 714 ms
9 SELECT * FROM airport
10 where airport_name LIKE '%Airport';
Services
Database
postgres@localhost
console_7 714 ms
console_5
Output
Result 15-2
Result 15-3
CSV
QUERY PLAN
1 Seq Scan on airport (cost=0.00..5.50 rows=1 width=57) (actual time=0.076..0.076 rows=0.00 loops=1)
2 Filter: ((airport_name)::text ~~ '%Airport'::text)
3 Rows Removed by Filter: 200
4 Buffers: shared hit=3
5 Planning Time: 0.169 ms
6 Execution Time: 0.097 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
10:36 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:34
09.11.2026


default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 create index idx_airport_country_city 37 ms
2 ON airport(country, city);
3
4 EXPLAIN ANALYZE 483 ms
5 select * from airport
6 where country = 'Philippines';
7
8 explain analyze 341 ms
9 SELECT * from airport
10 where city = 'Manila';
11
12 EXPLAIN ANALYZE 562 ms
13 select * FROM airport
14 where country = 'Philippines'
15 and city = 'Manila';
Services
Database
postgres@localhost
console_7 562 ms
console_5
Output
Result 16-2
Result 16-3
Result 16-4
CSV
QUERY PLAN
1 Seq Scan on airport (cost=0.00..6.00 rows=1 width=57) (actual time=0.071..0.071 rows=0.00 loops=1)
2 Filter: (((country)::text = 'Philippines'::text) AND ((city)::text = 'Manila'::text))
3 Rows Removed by Filter: 200
4 Buffers: shared hit=3
5 Planning Time: 0.179 ms
6 Execution Time: 0.115 ms
6 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
15:21 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:35
09.11.2026


default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 CREATE INDEX idx_lower_lastname 42 ms
2 on passengers(LOWER(last_name));
3
4 EXPLAIN ANALYZE 498 ms
5 select * from passengers
6 WHERE lower(last_name) = 'irnis';
7
8 explain analyze 611 ms
9 SELECT * FROM passengers
10 where last_name = 'Irnis';
Services
Database
postgres@localhost
console_7 611 ms
console_5
Output
Result 17-2
Result 17-3
Result 16-4
CSV
QUERY PLAN
1 Seq Scan on passengers (cost=0.00..1.04 rows=1 width=672) (actual time=0.044..0.044 rows=0.00 loops=1)
2 Filter: ((last_name)::text = 'Irnis'::text)
3 Rows Removed by Filter: 3
4 Buffers: shared hit=1
5 Planning Time: 0.574 ms
6 Execution Time: 0.077 ms
6 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
10:27 CRLF UTF-8 4 spaces
15°C
Cloudy
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default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
1 create INDEX idx_booking_cover 35 ms
2 on booking(passenger_id)
3 INCLUDE (ticket_price);
4
5 ANALYZE booking; 55 ms
6
7 EXPLAIN ANALYZE 574 ms
8 select ticket_price
9 FROM booking
10 where passenger_id = 100;
Services
Database
postgres@localhost
console_7 574 ms
console_5
Output
Result 17-2
Result 17-3
Result 18-3
CSV
QUERY PLAN
1 Seq Scan on booking (cost=0.00..1.02 rows=1 width=7) (actual time=0.018..0.018 rows=0.00 loops=1)
2 Filter: (passenger_id = 100)
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=15 read=1
7 Planning Time: 1.366 ms
8 Execution Time: 0.037 ms
8 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
10:28 CRLF UTF-8 4 spaces
15°C Cloudy
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default
Version Control
Database Explorer
DDL
postgres@localhost
booking
booking_flight
flights
columns 14
flight_id integer
sch_time text
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
created_at timestamp
console_7 [postgres@localhost]
Welcome to DataGrip
console_5 [postgres@localhost]
Tx: Auto
Playground
lab2.public
select airline_code, count(*) 423 ms
from airline
GROUP BY airline_code
having count(*) > 1;
-- қайталанған кодтарды түзету
WITH duplicates AS (63 ms
select airline_id, airline_code,
row_number() over (
partition by airline_code
order by airline_id
) as rn
from airline
)
UPDATE airline a
set airline_code = 'L7_' || a.airline_id
FROM duplicates d
where a.airline_id = d.airline_id
Services
Database
postgres@localhost
console_7 845 ms
console_5
Output
Result 19
lab2.pg_catalog.pg_indexes
Result 18-3
CSV
indexname
indexdef
1 airline_info_pkey CREATE UNIQUE INDEX airline_info_pkey ON public.airline USING btree (airline_id)
2 idx_unique_airline_code CREATE UNIQUE INDEX idx_unique_airline_code ON public.airline USING btree (airline_code)
2 rows
Database Consoles > postgres@localhost > console_7 [postgres@localhost]
26:29 CRLF UTF-8 4 spaces
15°C Cloudy
Поиск
ҚАЗ
23:36
09.11.2026